Step 1 - OUTPUT

When the user starts the grade calculator, the program displays:

- Pre-Class Activity marks (0 <= mark <= 100)
- Class Exercise marks (<= 0 mark <= 100)
- Test 1 mark (0 <= mark <= 100)
- Control Input: user enters ‘done’ to signal the end of the mark entry for each category

Step 2 - OUTPUT

The program should display:

- Prompt/Error messages: Input feedback/error messages for validation (out of range numbers or non numeric text)
- Category Averages & Entry Counts:
    - Pre-Class Activities average percentage and total entries.
    - Class Exercises average percentage and total entries.
    - Test 1 score percentage and total entries.
- Overall Midterm Mark: Calculated overall midterm percentage based on course weightings (PCA = 10%, CE = 20%, Test 1 = 70%)

Step 3 - Process

1. Initialize Data Structures:
    a. Create an array/list of category names:
        i. Pre-Class Activities, Class Exercises, Test 1
    b. Create empty lists “marks” (to store entered mark arrays) and “pcts” (to store category averages).
2. Collect Marks and Validate (Loop for each category):
    a. For each assessment category, initialize an empty list “cat_list”.
    b. Prompt the user for a mark or “done”.
    c. If entry equals “done”, break the input loop.
    d. Else (Validation):
        i. Convert entry to a floating-point number.
        ii. If mark is between 0 and 100 (inclusive), append it to “cat_list”.

iii. Else, display an out-of-range error message.
iv. If conversion fails (non-numeric text), display an invalid input error message.

3. Calculate The Category Averages:
    a. Append “cat_list” to “marks”.
    b. If “len(cat_list) > 0 “, calculate average as sum of marks/ number of marks and append to “pcts”.
    c. Else, append 0.0 to “pcts”.
4. Calculate Weighted Overall Grade:
    a. Compute overall percentage: all marks in each category/100

Step 4 - Pseudocode

```
START

CREATE empty list pre_class_marks
CREATE empty list class_ex_marks

PROMPT use to enter number of Pre-Class Activities
FOR each Pre-Class Activity
        PROMPT for mark
        IF mark invalid
                DISPLAY error
                REPEAT
        ADD mark to pre_class_marks

PROMPT user to enter number of Class Exercises
FOR each Class Exercise
        PROMPT for mark
        IF mark invalid
                DISPLAY error
        ADD mark to class_ex_marks

PROMPT for Test mark
IF Test mark invalid
```

    DISPLAY error
    STOP

CALCULATE Pre-Class average
CALCULATE Class Exercise average
CALCULATE overall midterm percentage

DISPLAY all results

END

Step 5 - Desk Check

Test Case Scenario

- Pre-Class Activities Inputs: 80, 90, ‘done’
- Class Exercise Inputs: ‘adc’ (invalid), 105 (out of range), 75, ‘done’
- Test 1 Inputs: 85, ‘done’